Revealing Hidden Decision Patterns in Machine Learning Models

Complete Project Documentation

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1. Project Overview

This project analyses and interprets the decision-making behaviour of self-supervised representation learning models in artificial intelligence. The central question is whether these models—specifically CLIP evaluated in a zero-shot classification setting—truly learn meaningful semantic patterns from data, or whether they exploit irrelevant features that accidentally correlate with the correct output .

Using Explainable AI (XAI) techniques (Grad-CAM and, optionally, Layer-wise Relevance Propagation), the project visualises model attention, quantifies spurious feature reliance, and presents findings through an interactive dashboard.

2. Problem Statement

Modern AI systems rely heavily on self-supervised and contrastive learning to extract features from large, unlabelled datasets. Impressive benchmark accuracy can mask a critical flaw: models may learn dataset shortcuts rather than the genuine concept behind the task. When deployed in high-stakes applications—medical diagnosis, autonomous vehicles, industrial inspection—this shortcut learning can produce dangerous failures.

This project provides a principled, reproducible investigation of shortcut learning by combining quantitative performance metrics with visual explanation methods.

3. Objectives

1. Analyse how self-supervised representation learning models (CLIP) build internal representations from image data.
2. Determine whether the model relies on semantically meaningful features or on irrelevant cues such as logos, text, or background artefacts.
3. Apply and compare Explainable AI techniques (Grad-CAM and LRP) to visualise and interpret model decisions.
4. Demonstrate the practical importance of interpretability for building reliable and trustworthy AI systems.
5. Quantify the extent of spurious-feature dependency across the ImageNet truck dataset.

4. Scope

The project focuses on analysing and interpreting decisions of image-based deep learning models using XAI. Work is limited to applying pre-trained models on the selected dataset and visualising model attention.

The project does NOT involve training new models or reproducing large-scale research experiments. The emphasis is on understanding decision-making processes and evaluating whether the model attends to semantically relevant image regions.

5. Models Used

5.1 Primary Model — CLIP

CLIP (Contrastive Language-Image Pre-training) learns visual representations by training on image–text pairs without explicit class labels. In this project it is evaluated in a zero-shot classification setting: a text prompt such as "a photo of a {class}" is compared against the image embedding, and the highest-similarity class is predicted. This makes standard classification metrics (accuracy, precision, recall, F1-score) both applicable and meaningful.

> ***Note:*** *CLIP's zero-shot capability is ideal for studying spurious correlations because the model was never directly optimised for truck classification—any reliance on logos or text is therefore a product of pre-training, not task-specific fine-tuning.*

5.2 Optional Comparison Models

- SimCLR — contrastive self-supervised model for potential accuracy/attention comparison.
- Barlow Twins — redundancy-reduction self-supervised approach for further comparison.

5.3 CLIP Variants to Test

Variant	Backbone	Notes
ViT-B/32	Vision Transformer	Default; requires transformer-compatible Grad-CAM (captum)
ViT-L/14	Vision Transformer (large)	Higher accuracy; more compute required
RN50 / RN101	Modified ResNet	Standard Grad-CAM works; fallback if ViT causes issues

6. Dataset

6.1 Primary Dataset — ImageNet Truck Subset

Size: ~2 GB | Source: torchvision / Kaggle / Hugging Face | Access: Public

Class Name	Synset ID	Spurious Feature Potential
Garbage truck	n03417042	Municipal markings, distinctive shape
Moving van	n03796401	Company logos on side panels
Pickup truck	n03930630	Brand emblems, common vehicle type
Trailer truck	n04467665	Text, branding on trailer body
Beer truck	n02814533	Prominent brand logos — PRIMARY test class
Fire engine	n03345487	Distinctive colour and emergency markings

Class Name	Synset ID	Spurious Feature Potential
Tow truck	n04461696	Unique mechanical equipment
Minivan	n03770679	Different body style, fewer logos

Rationale: trucks frequently display brand logos, company names, regulatory text, and distinctive backgrounds—ideal for investigating whether a model learns vehicle features or memorises logo patterns. Beer trucks and moving vans are prioritised for the Logo/Text Dependency Test due to their prominent spurious features.

7. Tools and Technologies

Tool	Purpose	License
Python 3.9+	Main programming language	Open source
PyTorch / torchvision	Deep learning framework & dataset loading	Open source
CLIP (OpenAI)	Pre-trained self-supervised model	MIT
torchcam / captum	Grad-CAM implementation (transformer support)	MIT
Zennit	LRP implementation (optional)	MIT
OpenCV	Image preprocessing & blurring	Apache 2.0
NumPy	Numerical computations	Open source
Pandas	Data manipulation & result logging	Open source
Plotly Dash	Interactive dashboard	MIT
Vercel	Dashboard deployment	Free tier
GitHub	Version control	Free
Google Colab	GPU compute (optional)	Free tier

8. Methodology

The project follows seven sequential stages:

Stage 1 — Dataset Preparation

Download the ImageNet truck subset (8 classes listed in Section 6). Write a PyTorch DataLoader that returns normalised images and integer labels.

Stage 2 — Model Setup

Load CLIP using the OpenAI library. Test both ViT-B/32 and a ResNet variant. Construct text prompts per class (e.g., "a photo of a {class}").

Stage 3 — Baseline Evaluation

Run zero-shot inference on the full test split. Record accuracy, precision, recall, and F1-score per class and overall.

Stage 4 — Image Modification

Create modified test images by: (a) blurring logo/text regions with Gaussian blur, (b) replacing those regions with average background colour, and (c) cropping to exclude logo areas where feasible. Re-evaluate the model on modified images to compute the Performance Delta.

Stage 5 — XAI Implementation

Apply Grad-CAM (via captum for ViT; torchcam for ResNet) to generate per-image attention heatmaps. Optionally apply LRP via Zennit. Overlay heatmaps on original images and save as PNG.

Stage 6 — Comparative Analysis

Quantify Attention Alignment Score (% of activation on truck body vs. background). Calculate Misleading Feature Dependency Rate. Compare Grad-CAM and LRP outputs for consistency. Produce side-by-side original vs. modified heatmap comparisons.

Stage 7 — Dashboard Development

Build a Plotly Dash application allowing users to upload images, select the model, run inference, and view heatmap explanations. Deploy on Vercel.

9. Stakeholders

Stakeholder	Description	Key Needs	Interest	Influence
Data Scientist	Develops & analyses models	Model interpretation, debugging tools, feature analysis	High	High
End User	Uses system outputs	Clear predictions, understandable heatmap explanations	High	Medium
Organisation	Deploys system in real applications	Reliability, scalability, safety, audit logs	High	High
Domain Expert	Validates model results	Accurate and meaningful explanations	Medium	Medium
Regulatory Body	Ensures compliance & ethics	Transparency, fairness, bias detection	Medium	High

10. Requirements

10.1 Functional Requirements

Note: *Requirements marked with [UC ref] map to the use cases in Section 11.*

FR-1 Dataset Handling

- The system shall load image datasets from disk, torchvision, or Hugging Face and apply standard preprocessing (resize to 224×224, normalise per CLIP spec). [UC6, UC7]
- The system shall support at least the eight ImageNet truck classes defined in Section 6. [UC1]

FR-2 Model Integration

- The system shall load the pre-trained CLIP model (ViT-B/32 as default; RN50 as fallback). [UC1, UC5]
- The system shall perform zero-shot inference using configurable text prompts. [UC1, UC4]
- The system shall optionally load SimCLR or Barlow Twins for comparison. [UC5]

FR-3 Model Evaluation

- The system shall compute accuracy, precision, recall, and F1-score for both original and modified test images. [UC7]
- Results shall be stored in a structured format (CSV) for reproducibility. [UC8]

FR-4 Explainable AI

- The system shall implement Grad-CAM (primary) and LRP (optional) to generate attention heatmaps. [UC2]
- The system shall overlay heatmaps on original images and save visualisations as PNG. [UC2, UC4]

FR-5 Result Analysis

- The system shall calculate the Attention Alignment Score for each image. [UC1, UC3]
- The system shall flag images where >30 % of attention falls on annotated spurious regions. [UC3]
- The system shall compute the Modified Image Performance Delta (original accuracy − modified accuracy). [UC1]

FR-6 Visualisation & Dashboard

- The system shall provide an interactive web dashboard (Plotly Dash) with image upload, model selection, and real-time heatmap display. [UC4]
- The dashboard shall display predicted class, confidence score, and a plain-language attention summary. [UC4]
- The dashboard shall be deployed and accessible via a public Vercel URL. [UC4]

FR-7 Multi-Method Comparison

- The system shall generate Grad-CAM and LRP heatmaps for the same image set and compute their spatial correlation. [UC2, UC5]

FR-8 Experiment Management

- All experiments shall be reproducible via fixed random seeds and versioned code on GitHub. [UC8]
- The system shall support saving and reloading experiment configurations. [UC8]

10.2 Non-Functional Requirements

NFR-1 Performance

- Inference time per image: < 2 seconds on GPU.
- Heatmap generation per image: < 5 seconds.
- End-to-end explanation (inference + heatmap + display): < 8 seconds.
- Dashboard initial load: < 3 seconds.
- GPU memory usage: ≤ 8 GB (compatible with free Google Colab tier).

NFR-2 Security

- Uploaded images shall not be persisted beyond the active session unless explicitly saved by the user.
- Role-based access control shall restrict administrative functions to authorised users.

NFR-3 Usability

- The dashboard UI shall require no ML expertise to operate.
- Heatmap overlays shall use a perceptually uniform colour map (e.g., jet or viridis) with a legend.
- Plain-language attention summaries shall be shown alongside every heatmap.

NFR-4 Reliability

- The system shall checkpoint experiment results after every batch to prevent data loss (mitigates Colab disconnections).
- Results shall be deterministic given the same random seed and input data.

NFR-5 Scalability

- The architecture shall be modular to permit addition of new models (e.g., DINOv2, MAE) and new XAI methods without major refactoring.

NFR-6 Maintainability

- Code shall be organised in clearly separated modules: data/, models/, xai/, dashboard/, evaluation/.
- All public functions shall include docstrings; the README shall document setup and reproduction steps.

NFR-7 Robustness

- The system shall handle corrupted or unusually shaped images with a graceful error message rather than a crash.
- Heatmap generation shall remain stable across CLIP ViT and ResNet backbones.

11. Use Cases

11.1 Use Case Summary

ID	Name	Primary Actor	Outcome
UC1	Analyse Model Decisions	Data Scientist	Heatmaps + interpretation report generated
UC2	Visualise Feature Importance	Data Scientist	Interactive heatmap gallery saved
UC3	Detect Bias / Spurious Features	Researcher	Bias report with flagged images
UC4	View Prediction with Explanation	End User	Prediction, confidence score, heatmap displayed
UC5	Compare Models	Data Scientist	Side-by-side performance and attention comparison
UC6	Train / Fine-tune Model	Data Scientist	Versioned model saved to repository
UC7	Test Model	Researcher	Metrics and XAI summary recorded
UC8	Generate Report	Organisation	PDF / HTML report exported
UC9	Monitor Model	Organisation	Live metrics dashboard with drift alerts
UC10	User Login	All roles	Authenticated session with role-based access

11.2 Detailed Use Case Descriptions

UC1 — Analyze Model Decisions

Field	Details
Type	Essential
Actors	Data Scientist, System
Pre-conditions	Logged in; dataset uploaded and validated; at least one pre-trained model available.
Post-conditions	Analysis report with heatmaps saved to workspace; session logged for audit.
Main Flow	1. Navigate to Model Analysis dashboard. 2. Select dataset and CLIP model. 3. Click Run Inference & Explainability. 4. System runs inference and computes metrics. 5. System generates Grad-CAM heatmaps and overlays them. 6. System flags images where attention is on spurious regions. 7. User inspects individual images or runs batch analysis. 8. System saves results (CSV + PNG heatmaps). 9. Summary dashboard shows % of images with object-focused vs. spurious-focused attention.
Alternatives	3a. Incompatible dataset → prompt to upload new one. 6a. Model not cached → download with progress bar. 10a. Weak attention signal → suggest LRP or ResNet backbone.

UC2 — Visualize Feature Importance

Field	Details
Type	Essential
Actors	Data Scientist, System
Pre-conditions	Model and dataset loaded (from UC1 session or fresh).
Post-conditions	Interactive heatmap gallery stored in project.
Main Flow	1. Open existing session or start new one. 2. Select XAI method (Grad-CAM or LRP). 3. System generates layered heatmaps. 4. Interactive gallery with zoom, layer toggle, raw/overlay switch. 5. Filter by confidence score or detected misleading focus. 6. Statistical summary (avg. attention foreground vs. background). 7. Save with annotations.
Alternatives	3a. Heavy model → Quick Preview mode (smaller sample).

UC3 — Detect Bias / Spurious Features

Field	Details
Type	Essential
Actors	Researcher, System
Pre-conditions	Model analysed on labelled / annotated dataset.
Post-conditions	Bias report generated; flagged issues highlighted.
Main Flow	1. Select Bias Detection from menu. 2. System scans metadata for environmental tags (logo present / absent, background type). 3. User defines bias criteria (e.g., performance drop on logo-heavy classes). 4. System runs subgroup analysis and calculates disparity metrics. 5. Bias heatmap and fairness scores generated. 6. Mitigation suggestions presented.
Alternatives	—

UC4 — View Prediction with Explanation

Field	Details
Type	Essential
Actors	End User, System
Pre-conditions	Logged in; at least one model deployed and ready.
Post-conditions	Prediction and explanation displayed; optionally saved to history.
Main Flow	1. Navigate to Quick Predict page. 2. Upload or drag an image. 3. System validates image and selects default model (CLIP ViT-B/32). 4. Inference + Grad-CAM applied. 5. Predicted class, confidence score, and heatmap overlay shown. 6. User can toggle between Grad-CAM and LRP. 7. Plain-language summary displayed (e.g., "Model focused 78 % of attention on the truck body").
Alternatives	—

UC5 — Compare Models

Field	Details
Type	Essential
Actors	Data Scientist, System
Pre-conditions	At least two models available in repository.
Post-conditions	Comparison report saved.
Main Flow	1. Select Model Comparison tool. 2. Choose 2–4 models. 3. Run same test dataset through all. 4. Unified dashboard: accuracy, F1, side-by-side heatmaps. 5. Sort by Meaningful Attention Score.
Alternatives	—

UC6 — Train / Fine-tune Model

Field	Details
Type	Essential
Actors	Data Scientist, System
Pre-conditions	Logged in with sufficient compute quota; dataset uploaded and pre-processed.
Post-conditions	Trained model versioned in repository; immediate XAI analysis offered.
Main Flow	1. Select Train New Model. 2. Choose architecture (SimCLR, Barlow Twins). 3. Set hyperparameters. 4. Live progress + GPU usage display. 5. Post-training evaluation and XAI analysis offered.
Alternatives	—

UC7 — Test Model

Field	Details
Type	Essential
Actors	Researcher, System
Pre-conditions	Model loaded; test dataset prepared.
Post-conditions	Metrics and XAI summary recorded.
Main Flow	1. Select model and test dataset. 2. Run inference; compute accuracy, precision, recall, F1. 3. Apply XAI to full test set. 4. Results in tabular and visual format.
Alternatives	—

UC8 — Generate Report

Field	Details
Type	Essential
Actors	Organisation, System
Pre-conditions	At least one analysis session completed.
Post-conditions	PDF or HTML report exported and stored.
Main Flow	1. Select one or more analysis sessions. 2. Auto-generate structured report (objectives, methodology, results, heatmaps, conclusions). 3. Customise sections. 4. Export.
Alternatives	—

UC9 — Monitor Model

Field	Details
Type	Essential
Actors	Organisation, System
Pre-conditions	Models deployed in production or analysis mode.
Post-conditions	Monitoring dashboard updated; alerts sent if drift or bias detected.
Main Flow	1. Open Monitoring Dashboard. 2. View live metrics, recent analyses, trend charts. 3. Alerts triggered if explanation quality drops below threshold.
Alternatives	—

UC10 — User Login

Field	Details
Type	Essential
Actors	All roles, System
Pre-conditions	User has a registered account.
Post-conditions	Authenticated; role-appropriate dashboard loaded; secure session token created.
Main Flow	1. Open application; click Login. 2. Enter email/password or use SSO. 3. System verifies credentials and account status. 4. Role assigned; personalised dashboard loaded.
Alternatives	3a. Invalid credentials → error message; locked after 3 failed attempts. 3b. Unverified account → email verification prompt.

11.3 User Stories

ID	Role	Story	Expected Outcome
US1	Data Scientist	As a data scientist, I want to analyse model decisions so that I can understand its reasoning behaviour.	Detailed attention maps + interpretation report
US2	End User	As an end user, I want to view predictions with visual explanations so that I can trust the system.	Increased confidence in model outputs
US3	Researcher	As a researcher, I want to detect spurious feature reliance so that I can propose improvements.	Quantified bias/shortcut evidence
US4	Organisation	As an organisation, I want a reliable, auditable system so that I can deploy it safely.	Safe, documented deployment
US5	Domain Expert	As a domain expert, I want to validate that heatmaps highlight correct image regions.	Validated and trustworthy explanations
US6	Data Scientist	As a data scientist, I want to compare multiple models so that I can select the most robust one.	Evidence-based model selection

12. Key Experiment — Logo / Text Dependency Test

12.1 Objective

Determine whether CLIP relies on brand logos and text overlays rather than truck-body features for zero-shot classification.

12.2 Procedure

6. Select 100 truck images with clearly visible brand logos (prioritise beer trucks and moving vans).
7. Measure baseline accuracy on original images.
8. Create modified versions using: (a) Gaussian blur on logo regions, (b) replacement with average background colour, (c) cropping to exclude logos where feasible.
9. Measure accuracy on modified images.
10. Generate Grad-CAM heatmaps for original and modified images.
11. Compare heatmap focus areas quantitatively (IoU with truck-body bounding box) and qualitatively.

12.3 Success Criteria

- Accuracy drop > 10 % → significant spurious feature reliance confirmed.
- Heatmap shift visible in > 50 % of original-vs-modified image pairs.
- At least 3 distinct spurious feature types identified (e.g., logos, background text, watermarks).

13. Evaluation Metrics (KPIs)

13.1 Predictive Performance

Metric	Description	Target
Accuracy	Overall correctness of zero-shot truck classification	> 80 % on test set
Precision	Of predicted class X, how many truly are X	> 75 % per-class average
Recall	Of all true class X instances, how many are identified	> 75 % per-class average
F1-Score	Harmonic mean of precision and recall	> 75 % overall

13.2 Attention Alignment Score

Measures the % of Grad-CAM activation falling within the annotated truck-body region (using bounding-box IoU where annotations are available; manual annotation otherwise).

Target: > 60 % of attention on the primary object.

13.3 Misleading Feature Dependency Rate

% of test images where > 30 % of attention falls on known spurious regions (logos, text, watermarks).

Target: Spurious dependency detected in ≥ 20 % of images (to demonstrate the effect exists).

13.4 Modified Image Performance Delta

(Accuracy on original images) − (Accuracy on modified images)

Delta > 10 % → significant spurious correlation. Delta < 5 % → relatively robust representations.

13.5 Cross-Method Consistency (XAI Robustness)

Spatial correlation between Grad-CAM and LRP heatmaps on the same image set.

Target: > 70 % agreement on primary attention regions.

13.6 Generalisation Gap

| Validation Accuracy − Test Accuracy |

Target: < 10 %. A large gap may indicate reliance on dataset-specific artefacts.

13.7 Qualitative Interpretability Assessment

Criterion	Scale	Description
Clarity	1–5	How clearly defined are highlighted regions?
Relevance	1–5	Does the highlight correspond to the actual object?
Consistency	1–5	Are similar images highlighted similarly?
Noise Level	1–5	How much scattered / random activation is present?

Target: Average score > 3.5 across all criteria (multi-rater evaluation).

13.8 Computational Efficiency

Metric	Target
Inference time per image	< 2 seconds
Heatmap generation time per image	< 5 seconds
Total explanation time (inference + heatmap + display)	< 8 seconds
Dashboard load time	< 3 seconds
GPU memory usage	≤ 8 GB (free Colab compatible)

13.9 Insight Generation

Criterion	Target
Documented failure cases	≥ 10 images where model focuses on spurious features
Spurious pattern types identified	≥ 3 distinct types (logos, background, text)
Side-by-side heatmap comparisons	≥ 5 original-vs-modified pairs
Actionable recommendations	≥ 3 concrete suggestions for improving robustness

13.10 Summary KPI Table

Metric	Type	Target	Priority
Accuracy	Quantitative	> 80 %	High
Precision / Recall / F1	Quantitative	> 75 %	Medium
Attention Alignment Score	Quantitative	> 60 % on truck body	High
Misleading Feature Dep. Rate	Quantitative	Detected in ≥ 20 % of images	Critical
Performance Delta	Quantitative	> 10 % drop = spurious	Critical
Cross-Method Consistency	Quantitative	> 70 % agreement	Medium

Metric	Type	Target	Priority
Generalisation Gap	Quantitative	< 10 %	Medium
Qualitative Interpretability	Qualitative	Avg > 3.5 / 5	Medium
Computational Efficiency	Quantitative	< 8 s total	Medium
Insight Generation	Qualitative	Meet all sub-targets	High

14. Task Assignment and Roles

14.1 Team Structure

Pair	Role	Person	Key Responsibilities	Est. Hours
Pair A	ML Engineering	Catherine	Load CLIP, preprocess images, run zero-shot predictions	25
Pair A	ML Engineering	Ahmed	Download ImageNet truck subset, write DataLoader, evaluation metrics, image modification	20
Pair B	XAI Integration	Jana	Implement Grad-CAM for CLIP (captum / torchcam)	25
Pair B	XAI Integration	Mariham	Implement LRP (Zennit), compare methods, heatmap overlay export	20
Pair C	Dashboard & Frontend	Hassan	Build Plotly Dash app, integrate model + XAI backend	35
Pair C	Dashboard & Frontend	Abdelrahman	Deploy on Vercel, write user docs, GitHub management	20

Total: ~145 hours over 14 weeks (~3–4 hours per person per week).

14.2 Pair A — Model & Data Engineering

Task ID	Description	Assignee	Deadline	Dependencies
A1	Set up Python environment; install all dependencies	Catherine	May 2	—
A2	Download ImageNet truck subset (8 synsets); organise into train/val/test splits	Ahmed	May 5	—
A3	Write PyTorch DataLoader (returns images and integer labels)	Ahmed	May 8	A2
A4	Load CLIP (ViT-B/32 and RN50); verify inference pipeline	Catherine	May 10	A1
A5	Write zero-shot prediction function (image → class label via text prompts)	Catherine	May 12	A4

Task ID	Description	Assignee	Deadline	Dependencies
A6	Write evaluation function: accuracy, precision, recall, F1 on test split	Ahmed	May 14	A3, A5
A7	Create modified images (blur/replace logos, alter backgrounds) for spurious feature testing	Ahmed	May 15	A3

14.3 Pair B — XAI Integration

Task ID	Description	Assignee	Deadline	Dependencies
B1	Install captum/torchcam; test Grad-CAM compatibility with CLIP ViT-B/32 and RN50 early	Jana	May 16	A5
B2	Implement Grad-CAM heatmap generation for a single image	Jana	May 22	B1
B3	Test Grad-CAM on original vs. modified images; document differences	Jana	May 25	A7, B2
B4	(Optional) Install Zennit; implement LRP for CLIP	Mariham	May 28	A5
B5	Compare Grad-CAM vs. LRP; select primary method for dashboard	Both	May 30	B3, B4
B6	Write function to overlay heatmap on original image and export as PNG	Mariham	May 30	B3

14.4 Pair C — Dashboard & Deployment

Task ID	Description	Assignee	Deadline	Dependencies
C1	Set up Plotly Dash project structure	Hassan	Jun 1	—
C2	Create UI: image upload, model selector (CLIP only), Explain button	Hassan	Jun 5	C1
C3	Integrate backend: call Pair A predictions + Pair B heatmaps	Hassan	Jun 10	A5, B3, C2
C4	Display heatmap and predicted class in dashboard	Abdelrahman	Jun 12	C3
C5	Add metrics display: attention alignment score, focus region analysis	Abdelrahman	Jun 15	C4
C6	Deploy dashboard on Vercel (free tier)	Hassan	Jun 18	C5
C7	Write user guide (README for dashboard)	Abdelrahman	Jun 20	C6

Task ID	Description	Assignee	Deadline	Dependencies
C8	End-to-end testing (protocol below)	Both	Jun 20	C7

Task C8 — End-to-End Testing Protocol:

12. Functional: upload 10 images per class (80 total); verify correct predictions.
13. XAI: verify heatmaps generate for all 80 images without errors.
14. Performance: measure average response time (target < 5 s per image).
15. Edge cases: rotated, cropped, and low-quality images.
16. UI: verify all buttons, dropdowns, and displays function correctly.
17. Deployment: confirm Vercel URL is accessible and stable.

14.5 All Pairs — Final Phase

Task ID	Description	Deadline
T1	Run Logo/Text Dependency Experiment on 100 logo-heavy images	Jul 5
T2	Collect results: class, heatmap, alignment score, spurious/correct focus per image	Jul 5
T3	Generate publication-quality figures (heatmaps, accuracy bars, before/after comparisons)	Jul 8
T4	Write final report (each pair contributes their section)	Jul 12
T5	Create presentation slides	Jul 14
T6	Rehearse presentation	Jul 16
T7	Final GitHub push with tag v1.0	Jul 17

15. Project Timeline

Interactive gantt chart -> click here

Phase	Tasks	Start	End	Duration	Owner
Planning	Proposal, Gantt, risk, KPIs	Apr 10	Apr 20	10 days	Pair A
Literature Review	Summarise 3–4 papers	Apr 10	Apr 20	10 days	Pair B
Requirements	Stakeholders, stories, functional/non-functional req.	Apr 10	Apr 20	10 days	Pair C
System Design	Architecture, data flow, UML diagrams	Apr 21	May 1	10 days	Pair A
Implementation — Model	Load CLIP, prepare ImageNet subset	May 2	May 15	2 weeks	Pair A

Phase	Tasks	Start	End	Duration	Owner
Implementation — XAI	Implement Grad-CAM / LRP for CLIP	May 16	May 30	2 weeks	Pair B
Implementation — Dashboard	Build Plotly Dash app, deploy on Vercel	Jun 1	Jun 20	3 weeks	Pair C
Testing & Evaluation	Experiments, heatmaps, metrics collection	Jun 21	Jul 5	2 weeks	All
Buffer / Contingency	Reserved for unexpected issues	Jul 6	Jul 10	5 days	All
Final Presentation	Slides, final docs, GitHub tag v1.0	Jul 11	Jul 17	7 days	All

15.1 Milestones

ID	Milestone	Due Date	Deliverable	Owner
M1	Planning & requirements docs complete	Apr 20	3 PDFs on GitHub	All pairs
M2	System design doc complete	May 1	PDF + UML diagrams	Pair A
M3	CLIP model loading + DataLoader ready	May 15	Working Jupyter notebook	Pair A
M4	XAI heatmap generation working	May 30	Heatmap example set (PNG)	Pair B
M5	Interactive dashboard deployed	Jun 20	Live Vercel URL	Pair C
M6	Experiments complete; results collected	Jul 5	CSV + heatmap images	All
M7	Final submission	Jul 17	GitHub repo tag v1.0	All

16. Deliverables

ID	Deliverable	Due Date	Format	Owner
D1	Project Planning & Management	Apr 20	PDF	Pair A
D2	Literature Review	Apr 20	PDF	Pair B
D3	Requirements Gathering	Apr 20	PDF	Pair C
D4	System Design Document	May 1	PDF	Pair A
D5	Source code (Jupyter + scripts)	Jul 17	GitHub	All
D6	Interactive dashboard (deployed)	Jul 17	Vercel URL	Pair C
D7	Final report (incl. results)	Jul 17	PDF	All

ID	Deliverable	Due Date	Format	Owner
D8	Presentation slides	Jul 17	PPTX / PDF	All

17. Risk Assessment and Mitigation

#	Risk	Prob.	Impact	Priority	Key Mitigation
R1	model learns shortcuts	High	High	Critical	Apply Grad-CAM + LRP; Logo/Text Dependency Test; diversify images
R2	Inconclusive or noisy XAI results	Medium	High	High	Use two XAI methods; tune visualisation params; document ambiguous cases
R3	Subjective evaluation of model behaviour	Medium	Medium	Medium	Define quantitative criteria (IoU, alignment score); multi-rater assessment
R4	CLIP under-performs on truck classification task	Low	Medium	Low	Test multiple CLIP variants and prompt formats before proceeding
R5	Grad-CAM incompatibility with ViT architecture	Medium	High	High	Use captum (transformer support); early testing in Week 1 of XAI phase; RN50 fallback
R6	Google Colab disconnections / GPU limits	Medium	Medium	Medium	Checkpoint to Google Drive after each batch; use small batch sizes
R7	Insufficient compute for large-scale inference	Low	High	Medium	Cloud GPU backup; smaller batch sizes; prioritise key experiments
R8	Model shows no clear spurious feature reliance	Medium	High	High	Test multiple classes; introduce controlled artefacts; document negative results
R9	Time constraints due to experiment complexity	Medium	High	High	Prioritise Logo/Text Test; pretrained models avoid training overhead; 5-day buffer
R10	Weak link between analysis and improvement proposals	Medium	Medium	Medium	Integrate improvement suggestions throughout; reference literature on mitigation
R11	Vercel deployment failure	Low	Medium	Low	Early deployment test by Jun 15; backup on Streamlit Cloud or local demo

18. Resource Allocation

18.1 Data Resources

Dataset	Size	Source	Access
ImageNet truck subset (8 synsets listed in Section 6)	~2 GB	torchvision / Kaggle / Hugging Face	Public

18.2 Software Tools

See Section 7 for the full tools table.

19. Conclusion

This project highlights the fundamental insufficiency of accuracy alone as a quality measure for AI systems. By evaluating CLIP in a zero-shot classification setting and applying Grad-CAM and LRP, the project reveals whether the model's impressive benchmark performance stems from genuine semantic understanding or from exploitation of spurious correlations—brand logos, background text, or dataset artefacts.

The structured combination of quantitative metrics (performance delta, attention alignment score, misleading feature dependency rate) and qualitative analysis (heatmap inspection, inter-rater assessment) ensures rigorous, reproducible conclusions. Findings will contribute to the broader understanding of self-supervised representation learning and support the design of more robust, interpretable AI systems.

Appendix A — Glossary

Term	Definition
Self-supervised learning	A paradigm where a model learns from unlabelled data by solving pretext tasks (e.g., predicting masked patches).
CLIP	Contrastive Language-Image Pre-training — learns visual representations from image–text pairs without class labels.
Zero-shot classification	Classification without task-specific training, using learned representations compared against text descriptions.
Grad-CAM	Gradient-weighted Class Activation Mapping — visualises model attention via gradients of the target class w.r.t. feature maps.
LRP	Layer-wise Relevance Propagation — back-propagates a relevance signal through network layers to highlight influential input regions.
Spurious correlation	A statistical correlation present in training data that does not reflect a true causal relationship.
Attention map / Heatmap	A colour-coded visualisation indicating which spatial regions of an input most influence the model's output.
Attention Alignment Score	Percentage of heatmap activation falling within the semantically relevant (truck-body) region of an image.
Performance Delta	Accuracy on original images minus accuracy on spurious-feature-modified images.

Appendix B — References

(To be completed during Literature Review phase — Deliverable D2)

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5. Zbontar, J., et al. (2021). Barlow Twins: Self-Supervised Learning via Redundancy Reduction. ICML 2021.

(Additional references to be added during literature review.)